Weekly Quant Radar

Prometheus Capital

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24/7 Markets: Implications for Quants, Part 3

® **Abstract:** Part 2 discussed the moves undertaken by Nasdaq and NYSE towards continuous markets. We focus on three immediate quant implications: the realised-variance corrections, the recurring perp-cash basis around maintenance pauses, and the overnight-vs-intraday anomaly that drives the measured equity premium. Each is examined with academic grounding to clarify what changes are expected to take place.

I. Volatility models like GARCH, HAR, and realised-variance decompositions treat intraday and overnight returns separately due to inherently different behaviour. Overnight returns have fatter tails, higher persistence, and are almost entirely feedback-driven [1], with extreme excess kurtosis relative to intraday returns. Under Nasdaq 23/5, "overnight" shrinks to a 1-hour technical pause. Jumps once packed into the close-to-open return now spread out into the continuous Night Session. The impact of the Hansen and Lunde [2] correction,

$$IV_c = c \cdot (RV + w \cdot | r_{CO} |^2),$$

becomes much smaller under Nasdaq 23/5: as the overnight window shrinks to a 1-hour pause, the scaling factor $c \to 1$ and the overnight weight $w \to 0$. It's shown [3] that the overnight share of total volatility has been rising from 1993 to 2017; under 23/5, that trend is expected to reverse. The natural replacement is a single continuous-vol process for the 23-hour session plus a small pause-gap correction, in the spirit of the GARCH-Itô framework [4], rather than the traditional open-to-close and close-to-open decomposition.

II. Nasdaq's 8:00–9:00 PM ET halt creates a recurring gap. QFEX perpetual futures trade through it. During the pause, adverse selection risk rises as in after-hours markets [5]. The no-arbitrage pricing of perpetuals [6, 7] implies that, during the pause,

$$P_{\text{perp}} = \mathbb{E}\left[P_{\text{spot}}^{\text{re-open}} \mid \mathcal{F}_{\text{pause}} \right] + \pi_{\text{funding}},$$

where π_{funding} is the funding-rate risk premium. In practice, this is a one-hour basis trade against the reopen auction – long (short) the perp at 8:00 PM ET against the cash open at 9:00 PM, with the funding payment as the carry. The opening auction's information-aggregation role [8] is correspondingly weakened, since only an hour of information has accumulated before each reopen.

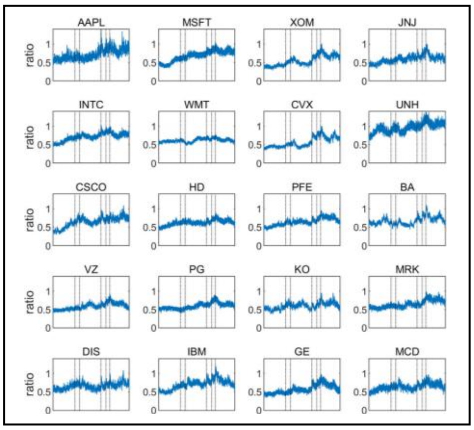
III. The separation of returns into close-to-open (r_{CO}) and open-to-close (r_{OC}) has been the bedrock of several high-profile anomalies. The "tug of war" hedge portfolio [9] earns roughly +3.5% p.m. on the overnight leg and loses about the same intraday, with the bulk of the measured equity premium concentrated in a single overnight hour driven by dealer inventory accumulation [10]. The split also shows up in betas: overnight CAPM betas are positive while intraday betas are negative [11]. Under 23/5, the overnight window effectively disappears. Whether these anomalies vanish entirely or transform into a new factor like,

$$r_{\text{pause-gap}} = P_{\text{re-open}} - P_{\text{pre-pause}},$$

will be an important empirical question this transition raises.

The launch of regulated 23/5 equity trading does not merely extend hours — it rewrites the variance, microstructure, and alpha-generation models that quants have relied on. Building the new ones is the next task.

Chart 1



Excerpt of rising ratios of overnight to intraday volatility, 1993-2017
Source: Linton & Wu, 2020, Journal of Econometrics

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